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# A multi-scale approach to hydrothermal liquefaction of two-phase olive mill waste: From laboratory test to regional biorefinery in Andalusia

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ABSTRACT

The olive oil industry produces over 3 Mt of two-phase olive mill waste (TPOMW) annually in Andalusia. Its management presents a significant environmental challenge due to its high moisture content and phytotoxicity. Hydrothermal liquefaction (HTL) is a promising valorization route utilizing the water content in the waste as a reaction medium, saving energy (no drying is needed). In this study a multi-scale approach has been used, integrating physicochemical characterization, laboratory scale experiments, kinetic modelling and a GIS-based location assessment for the location of HTL treatment plants. TPOMW shows high moisture (72 wt%) and a significant lipid content (16.4 wt% db), with a high heating value of approx. 23 MJ/kg. Kinetic results indicate that the solid fraction is the primary product for the first 22 min, after which the aqueous phase dominates due to rapid organic matter hydrolysis, obtaining a bio-oil yield around 20%. According to the resulting reaction network a model with three first order equation that can predict the behaviour of the process has been proposed. Finally, a GIS-based location-allocation analysis has been performed, proposing a decentralized network of 15 HTL plants across Andalusia. Results show a framework for the implementation of HTL-based biorefineries, supporting the transition toward a circular bioeconomy in the Mediterranean area.

## 1. Introduction

The olive oil industry is one of the most important economic activities in the Mediterranean area. About 95% of the world's olive growing is in Mediterranean area, including European Mediterranean coastal countries [1], being Spain the highest producer in the word. Specifically, the total area of olive groves in Spain was 2,671,278 ha in 2024 [2], with a production higher than 5 Mt of olive and nearly to 1 Mt of olive oil the year 2023. However, the oil industry generates large quantities of waste (wood, leaves, olive stones, olive mill waste, wastewater, etc.) producing several environmental problems [3], which are not adequately managed. The type of waste effluents obtained from olive oil industry depends on the production method [4,5]. In Andalusia, producing over 70% of Spain's olive oil, the two-phase system is widely used, accounting for over 90% of oil mills. This method was introduced to reduce water and energy consumption compared to the three-phase centrifugation [4]. It uses a decanter for two-phase centrifugation (oil and wet olive pomace) without water addition (see Fig. 1) and produces two streams: olive oil (around 20% of the fed) and a high-moisture solid residue, known as TPOMW. It is a highly problematic waste due to its high moisture content, the presence of phenolic, lipids and fatty acids compounds and its pronounced phytotoxicity. In Andalusia, the production is estimated to be higher than 3 Mt per year.

One of the most common ways to manage this waste has been by a solid-liquid extraction to obtain edible pomace oil followed by a combustion of the solid waste for electricity production [6]. However, this approach is declining due to technical and commercial reasons. From a

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**Glossary**

| | |
|---|---|
| CBO | Crude bio-oil |
| COP | Crude olive pomace |
| EVOO | Extra Virgin Olive Oil |
| HHV | Higher heating value |
| HTL | Hydrothermal liquefaction |
| ODEs | Ordinary differential equations |
| TPOMW | Two-phase olive mill waste |

technical perspective, it is inefficient as it is energy-intensive (i.e., drying the TPOMW prior to extraction), as well as costly (i.e., high transportation costs). From a commercial perspective, domestic as well as international consumers are favoring higher-quality olive oils (e.g., Extra Virgin Olive Oil: EVOO). Hence, olive oil production plants are facing the need to find alternative uses for TPOMW. These alternatives must align with the 2020 EU 'Circular Economy Action Plan', which aims to promote sustainable consumption and to reduce the amount of waste generated (also in in the agri-food system) by enhancing and re-evaluating waste products waste trying to reach the zero-waste target [7].

Nowadays, three different alternatives (routes) are being studied in literature: (1) the composting of the TPOWM to obtain an organic amendment to improve the quality and richness in nutrients in soils [8]; (2) the recovery of host bioactive compounds (as phenolic, minerals or polyunsaturated fatty acids compounds) and their further application in food, cosmetic, and pharmaceuticals [9,10]; and (3) the energy recovery by thermochemical conversion process, however, most of them have a high energy consumption in a dry step [11,12].

In this regard, HTL is a promising technology for producing a renewable liquid fuel from the TPOWM without the need for an energy-consuming drying process and reducing water consumption due to it uses the high moisture content of the feedstock to minimize the amount of water needed in the process. HTL has been proven as a suitable technology to convert feedstocks with high moisture content and heterogeneous composition into higher value products, obtaining mainly four fractions: an oily fraction (bio-oil), a solid fraction (hydrochar), an aqueous fraction, and a gas fraction [13]. Common raw materials are algae [14,15], biomass [16] or sewage sludge [17], although a few studies using wastes from the olive oil industry can be found [1,4,5,18,19]. However, the use of HTL for TPOMW is not sufficiently investigated. Filippis et al. [18] studied the HTL process for producing bio-oil from olive residue at 320 °C during 30 min, achieving yields ranging from 29% to 38%. Nonetheless, they did not specify the nature of the residue mixture used, although the characterization section indicates a moisture content of the used olive residue in the range 10-20%, ruling out the possibility that it was TPOMW. The same applies to Evcil et al. [1], who conducted HTL experiments in the temperature range of 250-330 °C and residence times of 5-60 min using an olive oil residue with a moisture content of 8.01%. Cantero-Tubilla et al. [19] focused on HTL to produce a hydrochar (the solid fraction) at two operating temperatures (250 and 300 °C) and times (5 and 60 min) from different waste feedstocks, including olive oil pomace, however, they did not provide data on the oil fraction. Dahdout et al. [4] demonstrated the feasibility of hydrothermal liquefaction of the two-phase olive mill wastes to produce bio-oil and hydrochar with high energy content. They performed a parametric study to screen different experimental operating conditions (a temperature range of 260-300 °C, water/dry-matter ratio of 20:1, 10:1 and 20:3 and residence times of 30-60 min). Nonetheless, they identified three outlet products with no aqueous phase; thus, the yield values were higher than expected. Moreover, they did not report any kinetic parameters. Finally, Cutz et al. [5] evaluated the influence of temperature (from 250 °C to 340 °C) on the bio-oil yield, with a residence time of 15 min. Moreover, they evaluated the impact of catalyst loading on improving both the yield and quality of the crude olive pomace (COP) bio-crude produced using HTL and examined the relationship between several operational variables. They also used a global kinetic model based on biomass composition to predict the yields obtained, but results were not satisfactory. Therefore, to the best of the authors' knowledge, no satisfactory model has yet been developed to reproduce the HTL process of TPOMW.

To evaluate the feasibility of this technology, this study follows a multi-scale approach structured into five main stages. First, the quantification, geographical location, and characterization of TPOWM generated in Andalusia were established using real production data. Second, HTL experiments were conducted to determine experimental yields, which were then used to propose and validate predictive models by comparing them with established literature data. Third, the process was further analyzed by developing a kinetic model and a reaction pathway to accurately represent the behavior of the process. Fourth, a biorefinery integration scheme was proposed and based on the obtained yields and waste availability, including a detailed mass balance for each product phase. Finally, optimization functions based on waste distribution were used to determine the optimal locations for the treatment plants, ensuring a sustainable and logistically efficient scale-up of the HTL process.

## 2. Material and methods

### *2.1. Quantification-location and characterization of two-phase olive mill waste (TPOMW). Availability of TPOMW in Andalusia*

TPOMW is the high-moisture solid residue obtained from the two-phase centrifugation method (Fig. 1). The TPOMW used in this study was provided by an oil extraction plant located in the province of Seville, Spain. The moisture content was determined by measuring the weight difference between the wet sample and the sample after drying in an oven at 105 °C until a constant weight was achieved. The content of volatile matter was performed according to standard UNE-EN

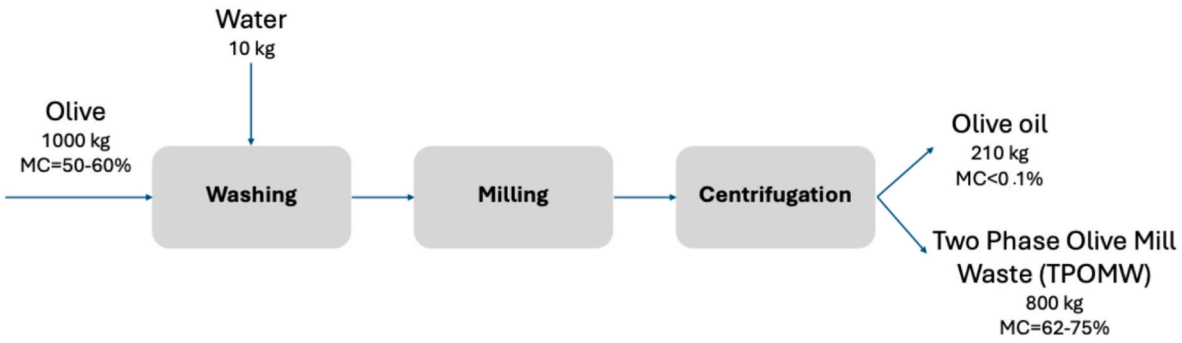



**Fig. 1.** Scheme of the two-phase system olive oil extraction method.

15148:2010, introducing the sample into the oven at 900 °C during 7 min, and determining the volatile matter by difference of weight. The ash content was quantified according to standard TAPPI T 211, after combustion for 1 h at 525 °C of 2 g of moisture-free sample basis. Proximate analysis was performed in quintuplicate to obtain the standard deviations (SD) associated. The determination of the content of fixed carbon was finally achieved by calculating the difference between the other components.

The biochemical composition and elemental analyzed were performed by an external laboratory of the moisture-free sample basis. The total lipid was calculated by gravimetry, the carbohydrate and protein contents were determined using the PNT.08.51 method and the elemental analysis using an Elemental Fison's Instruments EA 1108 CHNS. The higher heating value (HHV) of the feedstocks was calculated by Dulong equation (Eq. (1)):

$$\mathrm{HHV}\left(\frac{\mathrm{MJ}}{\mathrm{kg}}\right) = 0.338 \cdot C + 1.428 \cdot \left(\mathrm{H} - \frac{\mathrm{O}}{8}\right) + 0.095 \cdot S \quad (1)$$

where C, H, O and S are the carbon, hydrogen, oxygen and sulfur fractions, respectively, in the feedstocks, calculated on a dry basis.

To analyze the spatial distribution and quantity of available biomass, data on active olive mills in Andalusia were obtained from the Regional Government of Andalusia [20]. Additionally, information on waste generation from the olive oil industry in the region was collected from official regional reports. The correspondence between municipalities producing olive waste and those hosting olive mills was verified. In cases where municipalities reported TPOMW generation but lacked an olive mill, the waste amount was allocated to the location of the nearest olive mill. For municipalities containing multiple olive mills, a single representative point was used on the map, as intra-municipal distances were considered negligible. Waste availability and location data were integrated into an ArcGIS environment to facilitate a spatial analysis, which served as the foundation for performing a location-allocation for treatment plants based on waste density and transport logistics.

### 2.2. HTL experiments

HTL experiments were carried out in a batch reactor using 8 g of water and 2 g of dried sample (to keep constant a W/B ratio of 4:1 at different temperatures and residence times). The reactor used is a custom-built, Swagelok© stainless steel autoclave reactor with a Teflon-lined inner chamber was used to minimize corrosion. The reactor has a height of 10 cm, and a diameter of 3.5 cm. Fig. 2 shows the operational conditions employed in HTL tests. Tests at 300 °C and 325 °C for 60 min were performed by duplicity to study the reproducibility of experiments. In each run, the sample was placed inside the reactor, which was then sealed and heated to the desired temperature. Once this temperature was reached, the specified residence time was maintained to allow autogenous pressure, generated by the heated water, to develop in the system. After the selected residence time had elapsed, the reactor was allowed to cool to room temperature, after which the fractions were separated to determine the corresponding yields. The methodology and equipment setups for HTL runs and recovery steps of each fraction are described in our previous work [21].

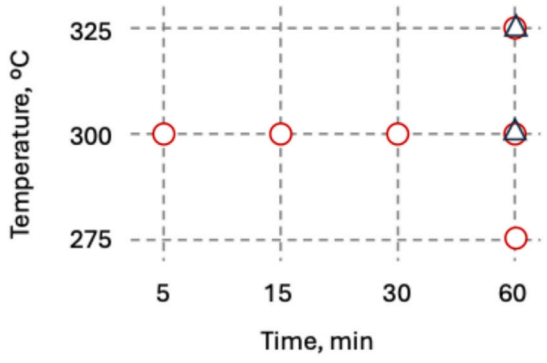



**Fig. 2.** Operational conditions employed in HTL experiments. Circles are initial tests and triangles the reproducibility ones.

### 2.3. Predictive and kinetic parameters models

To study the feasibility of the proposed process, the crude bio-oil (CBO) yield from TPOMW-HTL was predicted using different models based on TPOWM characterization data: (1) a linear predictive model based on lipid, protein, carbohydrate and ash fractions ($x_L$, $x_P$, $x_C$ and $x_A$) proposed by Li et al. [22] and (2) a quadratic model based on lipid, protein and carbohydrate fractions but considering also the biochemical interaction between these components, proposed by Teri and Savage, [23].

$$CBO\ yield_1\ (wt\%) = k_L x_L + k_P x_P + k_C x_C + k_A x_A \quad (2)$$

$$CBO\ yield_2\ (wt\%) = a x_L + b x_P + c x_C + d x_P x_C + e x_P x_L + f x_C x_L \quad (3)$$

where $k_L$, $k_P$, $k_C$ and $k_A$ in Eq (2) are the coefficient values for lipid, protein carbohydrate and ash, respectively, for different algae species at 300 °C and 30 min and $a$, $b$, $c$, $d$, $e$ and $f$ in Eq (3) are the parameters using cellulose, albumin and castor oil as feed under the same conditions and considering the biochemical interaction between the different components. CBO yields from obtained TPOWM characterization data were calculated using both models and they were compared with experimentally obtained at 300 °C and 30 min.

To determine the scheme reaction for HTL of the TPOWM, experimental data were adjusted to two reaction models based on reaction networks from literature for biomass, algae and lignocellulosic waste [5,16,24]. These models comprise a set of differential equations representing each reaction pathway within a batch reactor:

- Model 1: It is a simplified kinetic scheme commonly applied to lignocellulosic feedstocks, consisting of five primary reactions (Fig. 3a). Eqs (4)–(7) shows the corresponding equations of this model.

$$Solid: \frac{dx_1}{dt} = -(k_1 + k_2 + k_3)x_1 \quad (4)$$

$$Bio-oil: \frac{dx_2}{dt} = k_1 x_1 - k_4 x_2 \quad (5)$$

$$Aqueous\ Phase: \frac{dx_3}{dt} = k_2 x_1 + k_4 x_2 - k_5 x_3 \quad (6)$$

$$Gas: \frac{dx_4}{dt} = k_3 x_1 + k_5 x_3 \quad (7)$$

- Model 2: It is a more complex framework involving eight reactions that accounts for more secondary reactions and interactions between the resulting products. The reaction pathway is shown in Fig. 3b and corresponding equations in Eqs 8–11.

$$Solid: \frac{dx_1}{dt} = -(k_1 + k_2 + k_3)x_1 + k_4 x_2 \quad (8)$$

$$Bio-oil: \frac{dx_2}{dt} = k_1 x_1 + k_5 x_3 - (k_4 + k_6 + k_8)x_2 \quad (9)$$

$$Aqueous\ phase: \frac{dx_3}{dt} = k_2 x_1 + k_6 x_2 - (k_5 + k_7)x_3 \quad (10)$$

$$Gas: \frac{dx_4}{dt} = k_3 x_1 + k_7 x_3 + k_8 x_2 \quad (11)$$

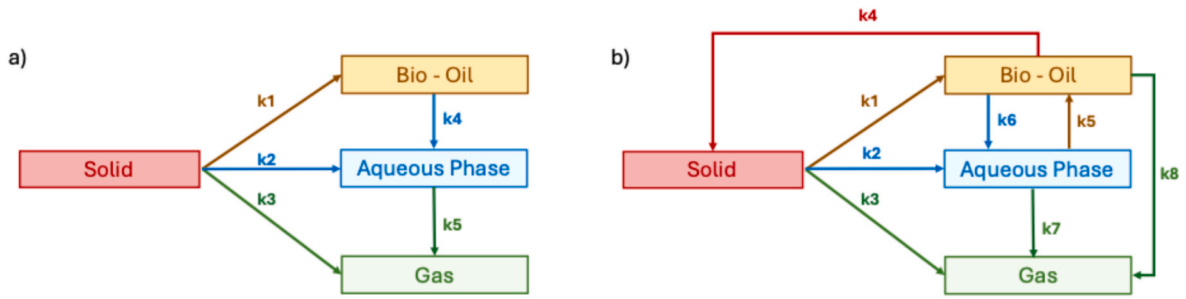



**Fig. 3.** Proposal reaction networks for HTL.

Experimental data were adjusted to these models and the specific kinetic parameters ($k_i$) for the used TPOMW were determined using experimental data and solving the system of ordinary differential equations (ODEs) using an explicit Rung-Kutta method. The kinetic rate equation for each reaction is assumed to follow first-order kinetics for the decomposition of biochemical components and the formation of biocrude, aqueous, and gas products. The results are further compared with relevant models from the literature.

### 2.4. Integration of obtained bio-oil in a biorefinery scheme

The obtained bio-oil by HTL processes can be used as a renewable substitute for petroleum-based feedstocks. However, it needs an upgrading step to be integrated into a biorefinery scheme. This involves the removal of moisture and ashes, and the reducing its density and heteroatoms contents (which is slightly higher than that of crude oil) [25]. Fig. 4 shows the biorefinery scheme proposed to optimize the utilization of all obtained fractions and upgrade them to produce a fully or nearly refined fuel for introduction into a refinery and the process balance was obtained according to obtained yields and literature data. In this scheme, solid phases and gases could be valorized in the plant (by a combustion unit) to obtain a heat stream which can be recirculated to THL reactor to partially supply the energy demanded by the process. Previous studies showed that the combustion of solid residue can supply 74% of the heat required by the HTL reactor [26]. Moreover, the aqueous phase could be conducted to an anaerobic digestion unit, to an environmental remediation and the production of a liquid soil amendment.

### 2.5. Location assessment for HTL treatment plants in Andalusia

Transport and siting analyses are critical for biomass-based systems, since feedstock logistics commonly represent a major share of both total system costs and greenhouse gas emissions [27]. To identify a suitable spatial configuration for HTL treatment plants in Andalusia, a location-allocation analysis was carried out in ArcGIS Pro using the regional road network [28]. TPOMW generation sites were represented as demand points weighted by the annual amount of residue generated at each olive mill, while candidate facility locations were defined at the same set of sites. Road-network distances were used instead of Euclidean distances to reflect actual transport routes.

The allocation problem was formulated as a capacitated location-allocation framework [29] implemented in ArcGIS Pro, in which the annual TPOMW generated at each olive mill $i$ ($b_i$) is assigned to selected HTL facilities $j$ through binary allocation variables $x_{ij}$. The objective is to maximize the total amount of biomass effectively allocated across the network (Eq. (12)).

$$\max \sum_{i\in I}\sum_{j\in J} b_i \cdot x_{ij} \tag{12}$$

Subject to capacity constraints at each facility. In particular, the total biomass assigned to each plant is limited by its maximum processing capacity $C$, set to 350 kt/a, such that:

$$\sum_{i\in I} b_i \cdot x_{ij} \le C \cdot y_j \quad \forall j \in J \tag{13}$$

where $y_j$ indicates whether a facility is established at location $j$. Demand

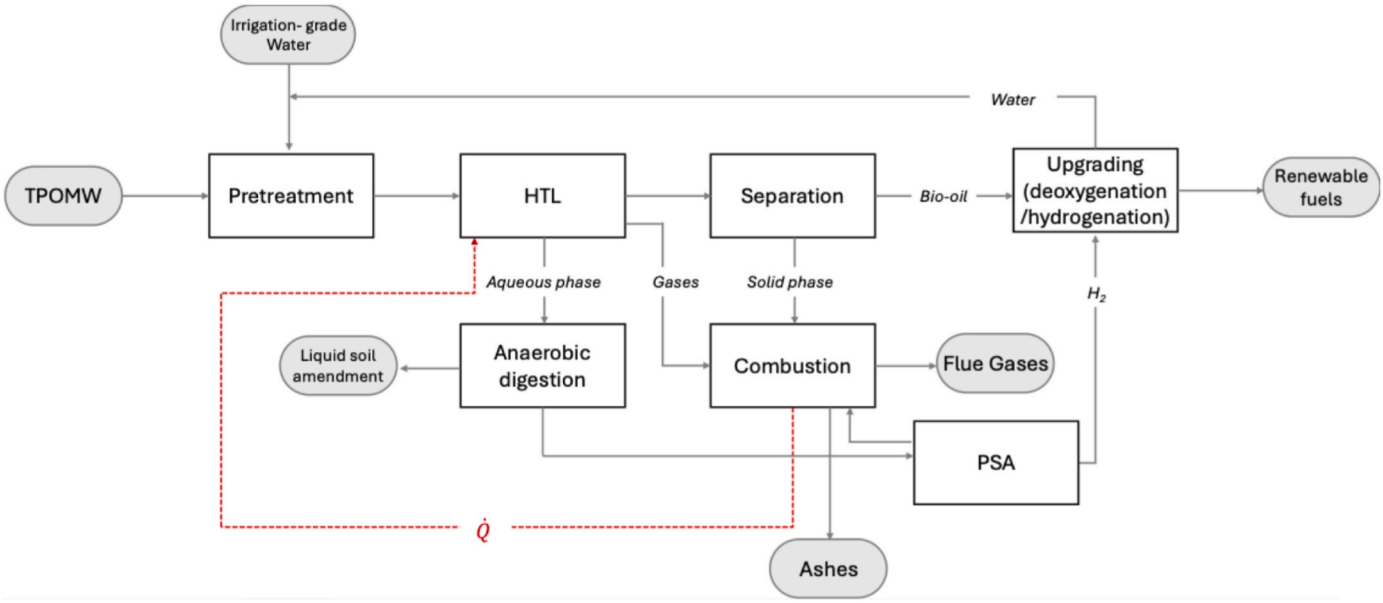



**Fig. 4.** Proposed biorefinery scheme for an optimal valorization of TPOMW.

nodes are treated as indivisible, meaning that each olive mill is assigned to a single facility.

A maximum processing capacity was assumed for each HTL facility to reflect realistic operational constraints and avoid excessively large, centralized plants. Based on typical scale ranges reported for thermochemical biomass conversion facilities in the literature, each HTL plant was assumed to process up to 350 kt per year of TPOMW [30]. Given the total regional biomass production, this constraint implies the need for a network of multiple facilities.

A first location-allocation run was performed to determine the spatial distribution of ten candidate HTL plants using the Maximize Capacitated Coverage problem type in ArcGIS, which allocates demand over the road network based on proximity while accounting for plant capacity constraints. Nevertheless, due to the discrete nature of the demand nodes, a small number of olive mills remained unassigned after the optimization. This occurs because demand points cannot be split among multiple facilities and therefore cannot be allocated when the remaining capacity of the nearest plants is insufficient to accommodate the full demand. To ensure that the entire regional TPOMW production was represented in the analysis, the remaining unassigned mills were subsequently allocated to the nearest selected HTL facility based on road-network distance. This additional step preserved the spatial configuration obtained from the GIS optimization while ensuring that the total biomass processed matched the total TPOMW generated in the region.

Road transport costs were estimated based on the proximity of each olive mill to its assigned HTL treatment plant, according to the final biomass allocation obtained from the GIS-based location-allocation analysis and the subsequent reassignment of unallocated mills. A unit cost of 1.3079 EUR/km was applied, as reported by the Spanish Ministry of Transport and Sustainable Mobility (July 2025) [31]. This cost corresponds to a general cargo articulated truck with a 25 t payload capacity. This factor includes both direct costs, as driver labor or maintenance and indirect costs. For each olive mill $i$, the total transport cost was calculated by Eq. (14), where $C_i$ is the transport cost for mill $i$ (EUR), $D_i$ is the distance from mill $i$ to the HTL site (km), $c$ is the cost per kilometre (EUR/km) and $n_i$ is the number of trucks required to transport TPOMW from mill $i$.

$$C_i = D_i \cdot c \cdot n_i \tag{14}$$

The specific transport cost for each olive mill (EUR/t) was obtained by dividing $C_i$ by the corresponding transported mass ($M_i$). These values were plotted as a function of transport distance to evaluate the spatial variability of logistics costs across the region. Moreover, the cumulative TPOMW generated was displayed on a secondary axis to indicate the potential processing capacity associated with increasing collection distance. Return trips were not considered, assuming empty backhauls, since the cost factor applied already integrates both time and distance-related operational expenses.

Road transport emissions were estimated for each olive mill based on the road distance to the assigned HTL treatment plant and the mass of TPOMW transported. Emissions per mill $i$ were calculated by Eq. (15), where $E_i$ is the emissions (kg $CO_2$eq), $D_i$ is the road distance from mill $i$ to the HTL site (km), $M_i$ is the mass of TPOMW transported (t), and $ef_{tkm}$ is the emission factor. A value of 0.1947 kg$CO_2$eq/(t·km) for the emission factor was applied, based on Ecoinvent entry Transport, freight, lorry 16-32 metric ton, EURO6 [32]. The specific emissions (kg$CO_2$eq/t) were then obtained by dividing $E_i$ by $M_i$. As with transport cost, return trips were not considered.

$$E_i = D_i \cdot M_i \cdot ef_{tkm} \tag{15}$$

## 3. Results

### 3.1. Characterization of TPOMW and availability in Andalusia

Table 1 shows results from characterization of the TPOWM used, as well as other TPOWM from the literature [4], [33–35]. The physicochemical characterization of TPOMW agrees well with this type of agro-industrial residue. The moisture content of the used sample in this work (72.36 wt%) was slightly higher than used TPOMW sample in literature, but it was aligned with the typical range of 55–75% for two-phase extraction systems, making this waste a highly suitable candidate for hydrothermal conversion processes. The fixed carbon and volatile matter contents on dry basis, 14.06 wt% and 74.53 wt% were the literature range (10.8-27.7% and 68.5-82.03%), while the ash content 11.41 wt% was significantly higher than literature values (3.2-7.2%), although it can be due to the specific harvesting conditions or regional soil characteristics. Regarding its biochemical composition, the lipid (16.4 wt%) and protein content (6.91 wt%) are consistent with the values reported for similar feedstocks in the Mediterranean region. It was noted that all samples from the literature had similar values for elemental analysis. Finally, the carbon content (57.82 wt%) of the waste and high heating value (23.47 MJ/kg) indicated the energy-dense nature of this waste. Overall, the characterization of TPOMW confirms its high potential for valorization through thermochemical or biochemical conversion routes [12].

The availability and location of TPOMW in Andalusia is shown in Fig. 5, categorized by annual tonnage per municipality. The analysis reveals a total annual production of 3,011,462 t, illustrating the vital importance of the olive oil sector across the entire region of Andalusia, as waste generation is observed in every province. This map shows that the provinces of Jaén and Córdoba have the highest biomass density, ranging from 18,001 to 64,000 t per year. The high amount of produced waste in Andalusia required a decentralized treatment strategy to manage the feedstock efficiently.

### 3.2. HTL experiments and predictive models

Operational conditions of Fig. 2 were tested in HTL experiments. Results are shown in Fig. 6. For duplicate experiments, average values were reported (it is noted that they had a standard deviation of less than 3.7% in all tests). Obtained yields for gas, solid, bio-oil and aqueous phase fractions at three tested temperatures at highest time (60 min) are shown in Fig. 6a). As the temperature increases, there is a sharp increase in the aqueous phase yield, reaching values of over 50% at 300 and 325 °C. The simultaneous decrease in the solid residue indicates that the thermal degradation of TPOMW promotes the solubilization of organic matter into the water-rich phase, resulting in a loss of the solid phase. This solubilized organic matter could be primarily composed of degradation products derived from cellulose and hemicellulose hydrolysis, lignin-derived water-soluble phenolics compounds, and proteinaceous and nitrogenous compounds originating from olive pulp [37]. The gas phase also increases with the temperature, but it is the minority fraction in the HTL process. The obtained gas phase yields were lower than 12% in all conducted tests. This fraction was not chemically analyzed in this work, literature on HTL of similar agricultural and biomass wastes indicated that this fraction is predominantly composed of $CO_2$ (typically exceeding 80% in volume) generated via decarboxylation reactions, accompanied by minor amounts of CO, $CH_4$, $H_2$ and light hydrocarbons [38,39]. Therefore, this fraction could be valorized using its sensible and heat combustion capacity. The bio-oil yield remains relatively stable across the tested range (around 25%). Fig. 6b) shows the effect of residence time at a fixed temperature of 300 °C. At the shortest reaction time of 5 min, solubilization of organic matter is mainly observed, while degradation of TPOMW practically does not occur (the solid fraction remains higher than 70% and the gas fraction is 0%). However, when the reaction time increases to 60 min, the solid fraction drops drastically

**Table 1**
Characterization of used TPOMW in this study and similar raw material from literature.

| | TPOMW | TPOMW | TPOMW | TPOMW | TPOMW | TPOMW |
|---|---|---|---|---|---|---|
| Reference | This study | [4] | [33] | [34] | [35] | [36] |
| Sample location | Seville, Spain | Ksar Kbir, Morocco | Manisa, Turkey | Jaen, Spain | San Juan, Argentina | Cordoba, Spain |
| Moisture (wt. %)[ar] | 72.36 ± 0.53 | 65.0 | 66.4 | 70.20 | 67.0 | 64.37 |
| Proximate analysis (wt. %)[db] | | | | | | |
| Volatile matters | 74.53 ± 0.86 | 80.5 | 68.5 | 76.62 | 82.03 | |
| Ash | 11.41 ± 0.15 | 3.2 | 3.8 | 5.32 | 7.18 | |
| Fixed Carbon[a] | 14.06 | 16.3 | 27.7 | 12.31 | 10.79 | |
| Compositional analysis (wt. %)[db] | | | | | | |
| Total carbohydrates | 7.42 | | | | | |
| Lipids | 16.4 | 12.4 | | | 13.37 | 1.39 |
| Protein | 6.91 | 7.1 | | | | 7.1 |
| Elemental analysis (wt. %)[db] | | | | | | |
| C | 57.82 | 51.2 | 51.47 | 57.82 | 51.25 | 48.15 |
| H | 7.05 | 6.9 | 7.02 | 7.02 | 7.57 | 7.14 |
| N | 0.65 | 0.9 | 1.09 | 0.65 | 0.92 | 1.18 |
| S | <0.10 | 0.3 | 0.13 | <0.1 | 0 | 0.0 |
| O | 34.43 | 39.2 | 36.51 | 34.43 | 31.4 | 43.53 |
| HHV (MJ/kg) | 23.47 | 20.2 | 26.1 | 23.65 | 18.61 | |

[ar] as received.
[db] on dry basis.
[a] by difference.

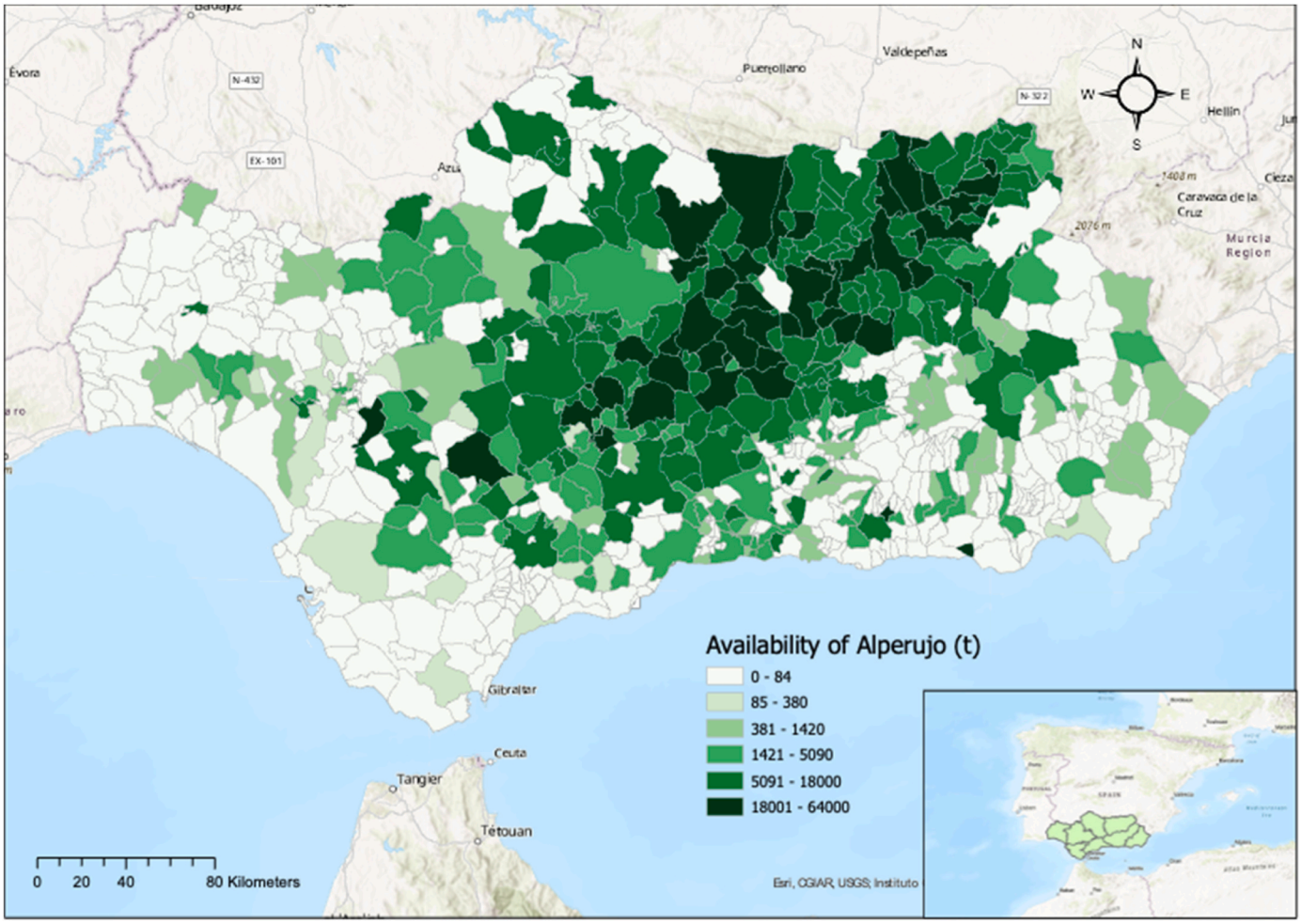



**Fig. 5.** Location and availability of TPOMW in Andalusia (t/y).

to around 12% (close to the ash content of TPOMW). Bio-oil yield increases progressively over time, reaching a maximum value of 25% after 60 min. This trend confirms that the conversion of TPOMW into liquid products via HTL is a time-dependent process in which secondary reactions play a crucial role in breaking down the initial solid matrix.

The experimental bio-oil yield obtained at 300 °C and 30 min

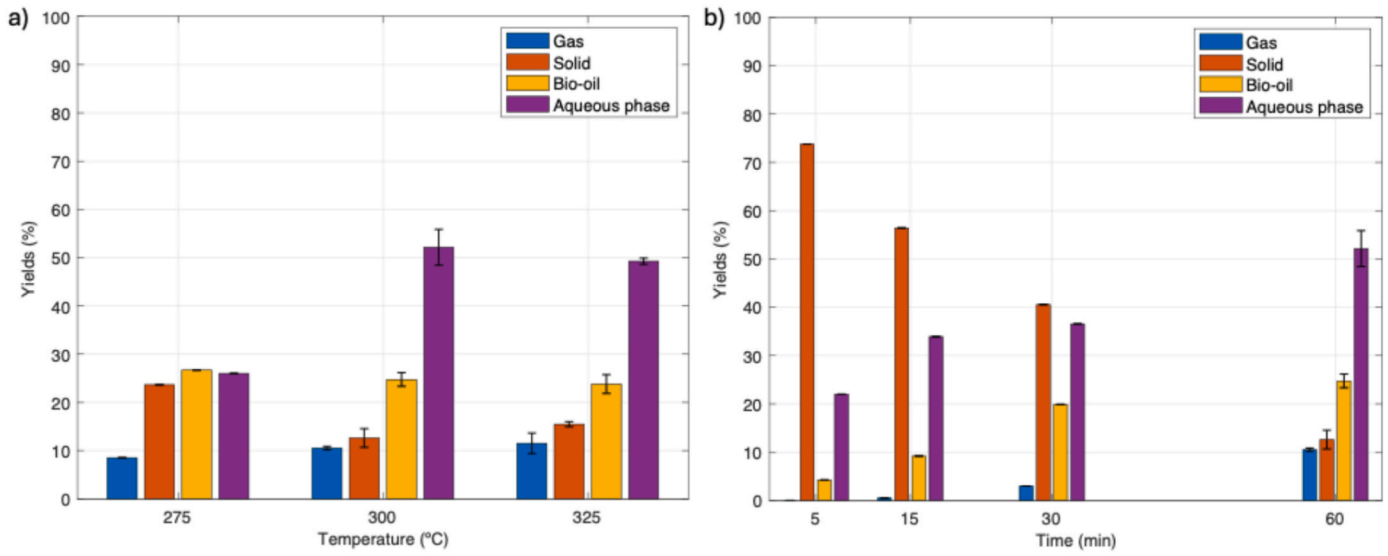



**Fig. 6.** Results from HTL experiments.

(19.9%) was compared with predictive models based on feedstock characteristics (Eqs. (4) and (5)). The estimated bio-oil yields from the data of Table 1 were 18.68% and 18.67% for the two models, respectively. The results indicate high model robustness since these models were originally obtained for other feedstocks [22,23], but they are remarkably applicable to TPOMW. Furthermore, the negligible difference between the linear and quadratic predictions suggests that the synergistic or antagonistic interactions between biochemical fractions do not significatively affect the bio-oil yield for TPOMW. Finally, a high agreement with experimental value was observed, indicating that the predictive models are adequate for our waste. However, these values are notably lower than reported by Cutz et al. [5], who achieved a value of 28.74% under non-catalytic HTL 300 °C and 30 min. This discrepancy can be attributed to the significant differences in the initial composition of the raw materials. Besides, the feedstock used had a lower ash and moisture content (3.6 and 57.2%, respectively) compared to the 11% ash and 72.36% moisture found in the TPOWM used in this study. A lower ash content implies a higher concentration of reactive organic matter, which can decompose into bio-oil during the HTL process. Moreover, these values also agree with the higher aqueous phase yield obtained in this work (versus a 22% obtained by Cutz et al. [5]), which is promoted by higher moisture content of the sample.

### 3.3. Modelling and kinetic parameters

To determine the HTL scheme reaction for TPOWM, two global kinetic models based on different reaction pathways (see Fig. 3) were adjusted using the experimental data to obtain the kinetic constants of the proposed reactions. It should be noted that in this study, the kinetic parameters were determined at the central temperature of 300 °C to evaluate the viability of the proposed reaction networks. Overall, the kinetic constants reported in Table 2 and the experimental data (black circles) together the corresponding kinetic model fits for all HTL fractions are showed in Fig. 7 (Model 1 as solid lines and Model 2 as dashed lines). Results provide a robust description of the system, with good agreement between experimental and modeled data, as shown in Fig. 7. The highest degree of agreement is observed for the solid and bio-crude fractions, while minor deviations are detected in the aqueous phase at short reaction times, where the model slightly underpredicts the initial decomposition rate. Both models, however, yield nearly similar fitting profiles.

**Table 2**
Kinetic constant for corresponding reactions of both proposed models.

| Kinetic constants *(1/s)* | Model 1 | Kinetic constants *(1/s)* | Model 2 |
|---|---|---|---|
| *k1* | $2.120 \cdot 10^{-4}$ | *k1* | 0.000 |
| *k2* | $4.824 \cdot 10^{-4}$ | *k2* | $7.381 \cdot 10^{-4}$ |
| *k3* | 0.000 | *k3* | 0.000 |
| *k4* | 0.000 | *k4* | $2.767 \cdot 10^{-4}$ |
| *k5* | $1.104 \cdot 10^{-4}$ | *k5* | $1.001 \cdot 10^{-3}$ |
| | | *k6* | $1.304 \cdot 10^{-3}$ |
| | | *k7* | 0.000 |
| | | *k8* | $1.472 \cdot 10^{-4}$ |

As observed in Table 2, some kinetic constant converged to 0 from the fitting models (*k3* and *k4* for Model 1 and *k1*, *k3* and *k7* for Model 2), indicating that the optimization algorithm identified them as negligible. Regarding *k3=0*, which represents the solid to gas reaction in both models, indicated that the direct decomposition of solid biomass into gas is extremely limited. Instead, gas formation predominantly occurs via secondary reactions, which is consistent with HTL mechanisms reported in the literature [5,14]. Similarly, *k4=0,* in Model 1 suggest a high stability of the formed bio-oil. For Model 2, *k1* and *k7* were also converged to 0. Regarding *k1=0,* it implies that this model considers the bio-oil does not form via a direct reaction from the solid, but that the process occurs through a consecutive mechanism (i.e., solid to water soluble intermediates in aqueous phase and subsequently reaction via condensation and repolymerization to produce the bio-oil fraction). Regarding *k7=0*, Model 2 suggest the gaseous products are predominantly derived from the secondary cracking of the bio-oil phase, leaving the *k7* pathway negligible. From these results, it can be concluded that, in a simpler reaction pathway, the reactions from the solid fraction to the gas phase and from the bio-oil to the aqueous phase are negligible. According to Model 2, it can be assumed that the solid decomposes to form an aqueous phase, from which the bio-oil fraction is formed. Three reactions then take place within the bio-oil fraction. The reaction networks obtained from both models are shown in Fig. 8.

Future work will focus on apply this kinetic framework to analyze the effect of temperature on kinetic constants via the Arrhenius equation, allowing for the determination of activation energies across the entire operational range.

Fig. 9 shows the comparison between the proposed kinetic models and experimental data for the HTL of TPOMW at 300 °C. The results indicate that the solid fraction is the primary product at residence times shorter than 22 min. However, as the reaction progresses, the aqueous phase becomes the dominant fraction due to the rapid hydrolysis and decomposition of the biomass organic matter. As the proposed kinetic

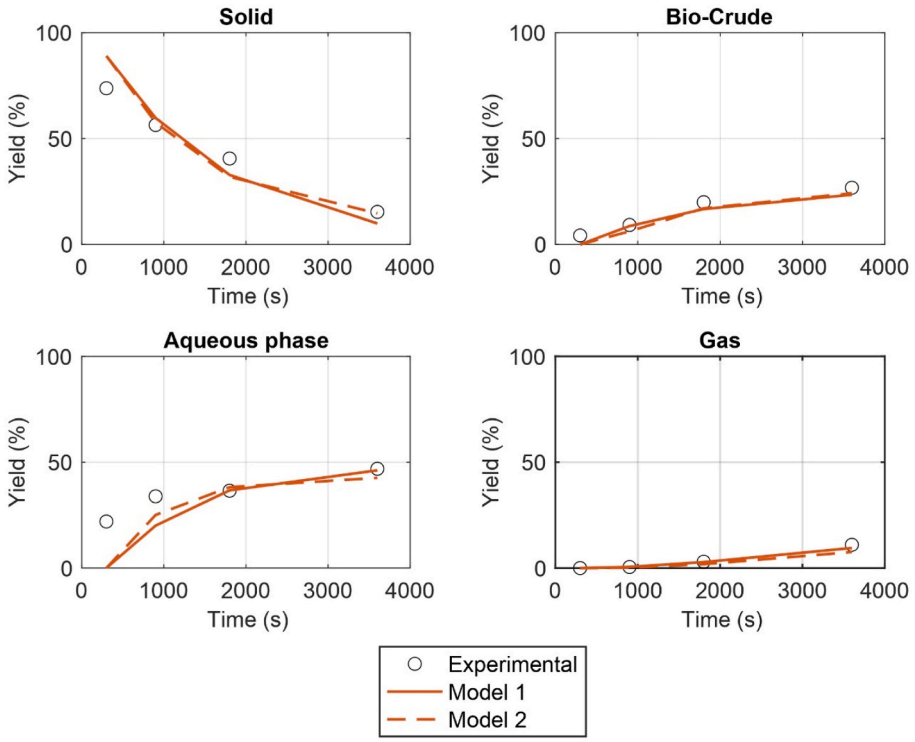



**Fig. 7.** Experimental and modelling data for both proposed models.

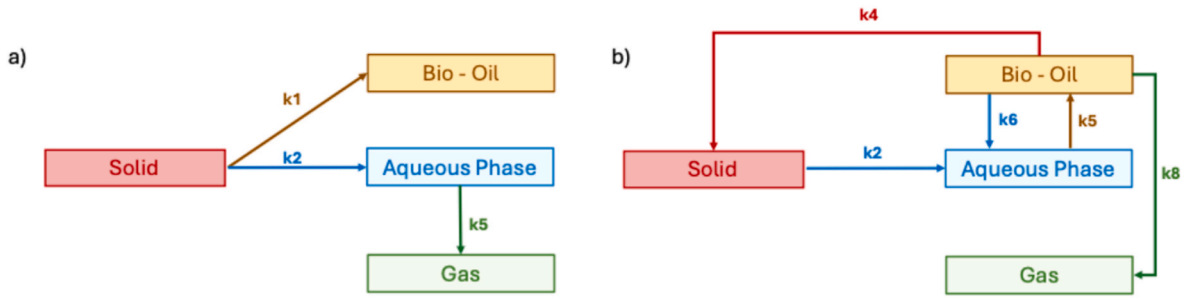



**Fig. 8.** Obtained reaction network from both proposed models for HTL.

model shows, the system approaches a stabilization after near to 60 min, particularly for the main bio-oil and aqueous fractions, indicating that longer residence times would not significantly alter product distribution. The gas phase remains the product with the lowest mass yield, as it is generated through secondary degradation pathways of the bio-oil and aqueous phase intermediates, characterized by relatively small reaction rate constants. It can be observed that both models accurately describe the process and both show a similar trend. In this sense, although Model 2 represents a more complex reaction pathway considering secondary interactions, experimental results are also well fitted with the simpler model (i.e., with only three reactions). However, final trend sat 60 min are subtly better adjusted with Model 2, showing a higher yield for solid fraction (13.54% versus 7.31%) and lower yields for bio-oil fraction (24.35% versus 24.94%), aqueous phase fraction (42.45% versus 43.37%) and gas fraction (8.65% versus 13.38%).

Finally, the proposed models and kinetic constants were used for similar residues [5,14] to predict the fraction yields, based on the composition of the feedstock (Table 1). However, significantly higher bio-oil yield values were observed (34.79% and 39.30% for both models, respectively) than in the experimental data at 300 °C for 30 min. For the model proposed by Valdez et al. [14], this may be because their model was developed for algae with much lower ash content. However, Cutz et al. [5] proposed a model for a crude olive pomace HTL process (referred to as *alpeorujo* in their work), but their raw material had lower moisture and ash content (57.2 and 3.6%, respectively) than the TPOMW used in this study. Therefore, the results cannot be directly compared, as the raw materials were different.

### 3.4. Location assessment for HTL treatment plants in Andalusia according a biorefinery scheme

Firstly, a GIS-based location-allocation analysis identified a network of ten suitable HTL plant locations across Andalusia, minimizing the weighted transport distance between TPOMW generation sites and processing facilities. Because the capacitated location-allocation model does not allow demand nodes to be divided between facilities, a small number of olive mills remained unassigned after the initial optimization. These mills were subsequently allocated to the nearest selected plant based on road-network distance, ensuring that the total TPOMW generated in the region was fully represented in the analysis. The final biomass processed by the selected plants ranges from approximately 70 to 380 kt/y. Overall, the selected network of facilities enables the processing of the entire regional TPOMW generation while maintaining a balanced spatial distribution of biomass supply. Nevertheless, in this

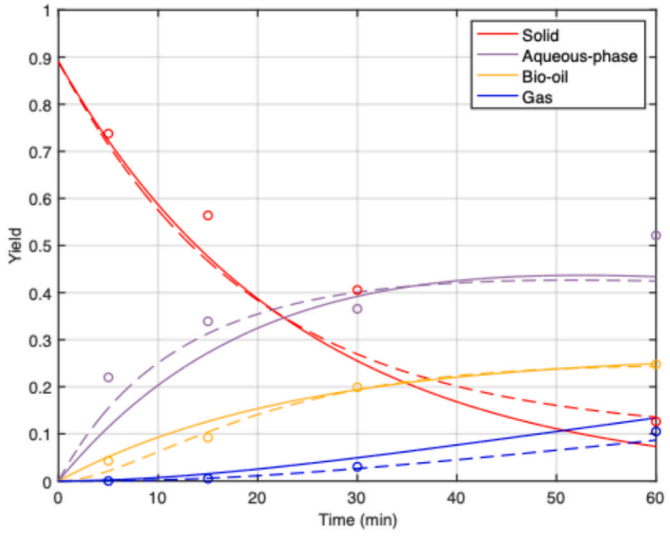



**Fig. 9.** Kinetic model results for TPOMW HTL at 300 °C using both proposed models: Model 1 (continuous line), Model 2 (dashed lines) and experimental data (circles).

configuration, some plants exceeded 350 kt/a, which may be unrealistic given the current level of commercial development of HTL technology. Moreover, the selected facilities were predominantly concentrated in eastern Andalusia, resulting in maximum transport distances above 150 km for a significant share of the biomass, which might not be suitable for wet residues such as TPOMW. For these reasons, the number of facilities was increased to fifteen to obtain a more feasible and spatially balanced configuration. The distribution of TPOMW processed annually by each plant with the fifteen plants configuration is summarized in Table 3.

Fig. 10 presents the study area, showing the distribution of TPOMW generation sites and the locations of the ten HTL plants selected through the GIS analysis. Olive mills are represented as proportional markers according to the amount of residue generated, while the selected plant locations are highlighted within the regional road network. Transport distances were calculated along the road network between each olive mill and its assigned HTL facility. The resulting transport distances ranged from short local deliveries to longer interregional transport routes, resulting in a maximum road distance of 133 km and an average distance of 41 km.

Fig. 11 shows the variation of specific transport cost and specific transport emissions with distance from the optimal HTL site. Both variables increase progressively with distance, but three distinct trends can be observed, reflecting changes in the marginal contribution of distance to overall transport performance.

Zone I (≤10 km) represents the most efficient collection area, where both cost and emissions increase almost linearly with distance. Within this range, the average specific transport cost is around 0.20 EUR/t and the mean emissions are approximately 0.73 $kgCO_2eq/t$, corresponding to the lowest logistical and environmental impact. With a transport distance of up to 10 km, the amount of TPOMW processed at the plant could be up to approximately $5.10{\cdot}10^5$ t. Zone II (10-65 km) shows a moderate rise in both indicators, reaching mean values of 1.84 EUR/t and 6.84 $kgCO_2eq/t$. This range marks the onset of declining efficiency, as longer transport distances reduce the economic attractiveness of biomass delivery. For this zone of the graph, the total amount of waste received at the plant would be $2.75{\cdot}10^6$ t for a supply distance of 65 km. Zone III (>65 km) displays a marked increase in both cost and emissions, exceeding 4 EUR/t and 16 $kgCO_2eq/t$, respectively. Beyond this distance, the transport of TPOMW becomes progressively less viable due to the decreasing ratio between transported mass and distance.

These three regimes delineate the practical limits for cost-effective road transport of TPOMW within the regional supply network. In contrast to a centralized configuration, in which a single facility defines a single supply radius, HTL's distributed system of ten plants creates multiple zones, thereby reducing the average distance travelled. The results obtained for the transport stage indicate that costs and emissions increase sharply beyond approximately 200 km. This distance can therefore be interpreted as an upper bound for economically and environmentally viable transport within the regional system, rather than as a strict supply radius for an individual facility. Literature on biomass logistic supports this finding, reporting that road transport is typically competitive only up to distances ranging from 50 to 386 km, with commonly cited efficient collection radii in the range of 50–75 km [39]. In this context, the present results indicate that most of the TPOMW is supplied within moderate transport distances, while long-distance transport (<200 km) represents a marginal and less efficient contribution to the overall system.

Finally, according to the processed capacity by each plant of Table 3, the average value of obtained biocrude yields from TPOMW at 300 °C and 60 min for both models (24.65%), the moisture content (72%) and the proposed biorefinery scheme for an optimal valorization of TPOMW (shown in Fig. 4), the amount of renewable biocrude could be produced by each HTL plant was estimated and results are summarized in Table 4. Balances were determined according to predicted results of Model 1, with yields of 13.54%, 24.35%, 42.45% and 8.65% for solid, bio-oil, aqueous and gas fractions, respectively. It was considered that the proposed biorefineries work 365 days per year for 24 h. Respect the recuperation of solid fraction, based on localized plant capacities, the solid fraction productions would be ranged between 1000 and 15,180 t per year. This volume allows for the implementation of standard industrial separation units to continuously feed the downstream combustion chamber proposed in the biorefinery plant. Other critical factors of the biorefinery scheme were also estimated, such as the raw water demand (assuming a water quality equivalent to irrigation-grade water) as well as the production of ashes and liquid soil amendment. It should be noted that the irrigation water requirement for 1 ha of intensive olive grove ranges between 2410 and 4720 $m^3$/year [40], yielding an average of 10 t of olives per hectare. Consequently, the raw water consumption of the HTL process per t of treated TPOMW is less than 0.2% of the volume of water required in irrigation to produce 1 t of TPOMW. A significant generation of residual ashes was observed, which is directly attributed to the high ash content of the initial TPOMW (higher than 11 wt% db). Finally, the resulting liquid soil amendment could be applied to the surrounding olive groves, enhancing soil quality and promoting a circular economy within the local agricultural sector.

Based on the bio-oil production estimates presented in Table 4, the energy potential of the of the bio-oil produced at each HTL facility was assessed. Assuming a HHV of 29.5 MJ/kg, corresponding to bio-oil obtained from HTL of TPOMW under comparable conditions [4], the total

**Table 3**
Distribution of processed TPOMW across each HTL treatment plants proposed in the study.

| HTL Plants | Processed TPOMW (kt) | Latitude | Longitude | Province |
|---|---|---|---|---|
| #1 | 345 | 37.52 | −4.65 | Cordoba |
| #2 | 235 | 37.89 | −4.38 | |
| #3 | 305 | 37.51 | −4.15 | |
| #4 | 82 | 37.49 | −2.77 | Granada |
| #5 | 309 | 37.43 | −3.68 | |
| #6 | 45 | 36.96 | −3.05 | |
| #7 | 205 | 38.17 | −3.77 | Jaen |
| #8 | 164 | 38.25 | −3.13 | |
| #9 | 345 | 37.75 | −3.91 | |
| #10 | 22 | 37.87 | −4.19 | |
| #11 | 133 | 38.35 | −2.80 | |
| #12 | 295 | 38.01 | −3.37 | |
| #13 | 205 | 37.00 | −5.02 | Malaga |
| #14 | 126 | 36.93 | −4.19 | |
| #15 | 150 | 37.39 | −6.00 | Seville |

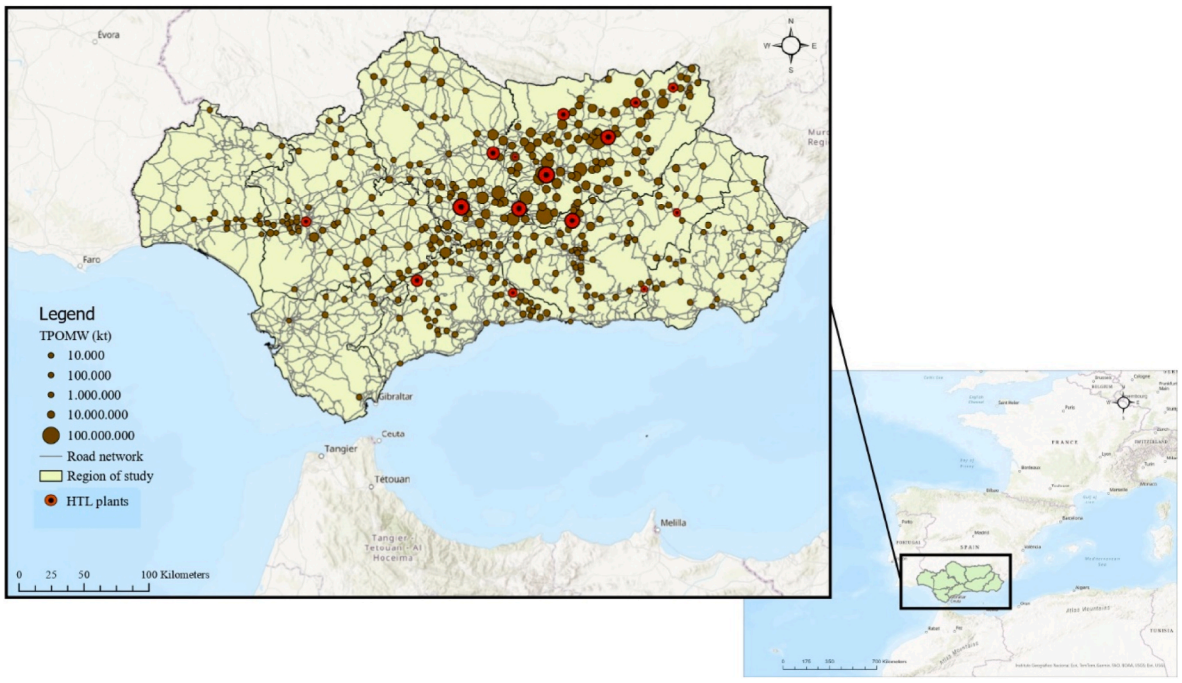



**Fig. 10.** Study area showing TPOMW generation sites, road network, and optimal HTL plant locations.

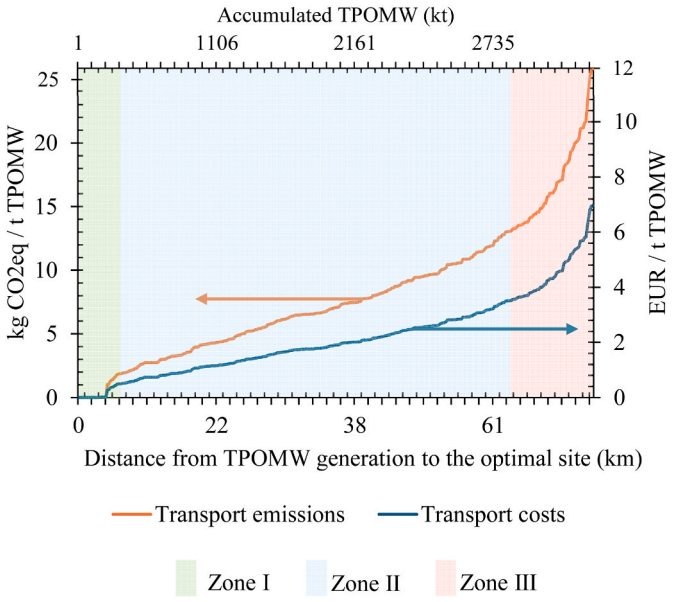



**Fig. 11.** Specific transport cost (EUR/t) and specific transport emissions (kg $CO_2$eq/t) as a function of total road kilometers from TPOMW generation to the HTL site.

energy output of each facility was calculated. The results, presented in Fig. 12, show the distribution of bio-oil energy potential across the proposed network of HTL plants.

Bio-oil was selected as the reference product for the energy assessment, rather than comparing its performance with the direct energy recovery from untreated TPOMW. This approach is justified by the practical limitations associated with the direct use of untreated TPOMW for electricity generation. The high moisture content of this waste significantly reduces combustion efficiency while increasing CO emissions [41]. In addition, the current management of TPOMW in Andalusia supports the need for alternative valorization pathways. Olive mills currently incur management costs of approximately 6 EUR/t for TPOMW disposal, while limited storage capacity can disrupt mill operations during peak processing periods [42]. Under these circumstances, HTL conversion into bio-oil represents a more practical and sustainable pathway for the regional valorization of TPOMW.

## 4. Conclusions

Firstly, the characterization demonstrated that used TPOMW exhibits a reliable chemical profile for thermochemical valorization pathways. The laboratory-scale experiments successfully validated both proposed kinetic mechanisms and the mathematical fitting revealed that both reaction networks effectively represent the process. However, the Model 1 showed a simplified network, which is preferred for the scaling up. Furthermore, the proposed biorefinery scheme proves to be highly efficient by utilizing the intrinsic moisture of the raw TPOMW as the reaction medium, which reduces fresh process water requirements (the proposed process requires less than 0.2% of the water volume typically used for the irrigation of each ton of TPOMW produced) and valorizing all obtained fractions. Finally, a decentralized biorefinery case study was implemented for Andalusia, where 3 Mt of TPOMW are generated annually. Through ArcGIS optimization aimed at minimizing transportation costs and carbon footprints, 15 strategic treatment plants were identified across Andalusia. These facilities allow producing in Andalusia 155,341 t of biofuel per year, with individual capacities ranging from 22 to 345 kt of TPOMW per year, were located using ArcGIS to minimize transportation costs and energy requirements while maximizing feedstock accessibility in high-production provinces like Jaen and Cordoba. Each plant is estimated to produce between 130 and 2062 kg/h of renewable biofuel. Lastly, this regional deployment confirms that combining a HTL process with optimized logistics for hubs allocation offers a highly sustainable alternative for seasonal agricultural wastes.

## CRediT authorship contribution statement

**A. Ronda:** Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **S. Moreno:** Visualization, Software, Resources, Methodology, Formal

**Table 4**
Distribution of processed TPOMW across each HTL treatment plant proposed in the study.

| HTL Plants | Processed TPOMW (kg/h) | Processed TPOMW ($kg^{db}$/h) | Raw water (kg/h) | Final ashes (kg/h) | Liquid soil amendment (kg/h) | Bio-oil (kg/h) | Renewable biofuel (kg/h) |
|---|---|---|---|---|---|---|---|
| Plant 1 | 39,384 | 11,027 | 15,616 | 1258 | 29,335 | 2750 | 2063 |
| Plant 2 | 26,826 | 7511 | 10,637 | 857 | 19,982 | 1873 | 1405 |
| Plant 3 | 34,817 | 9749 | 13,805 | 1112 | 25,934 | 2431 | 1824 |
| Plant 4 | 9361 | 2621 | 3712 | 299 | 6972 | 654 | 490 |
| Plant 5 | 35,274 | 9877 | 13,986 | 1127 | 26,274 | 2463 | 1847 |
| Plant 6 | 5137 | 1438 | 2037 | 164 | 3826 | 359 | 269 |
| Plant 7 | 23,402 | 6553 | 9279 | 748 | 17,431 | 1634 | 1226 |
| Plant 8 | 18,721 | 5242 | 7423 | 598 | 13,945 | 1307 | 981 |
| Plant 9 | 39,384 | 11,027 | 15,616 | 1258 | 29,335 | 2750 | 2063 |
| Plant 10 | 2511 | 703 | 996 | 80 | 1871 | 175 | 132 |
| Plant 11 | 15,183 | 4251 | 6020 | 485 | 11,309 | 1060 | 795 |
| Plant 12 | 33,676 | 9429 | 13,353 | 1076 | 25,084 | 2352 | 1764 |
| Plant 13 | 23,402 | 6553 | 9279 | 748 | 17,431 | 1634 | 1226 |
| Plant 14 | 14,384 | 4027 | 5703 | 460 | 10,714 | 1004 | 753 |
| Plant 15 | 17,123 | 4795 | 6790 | 547 | 12,755 | 1196 | 897 |

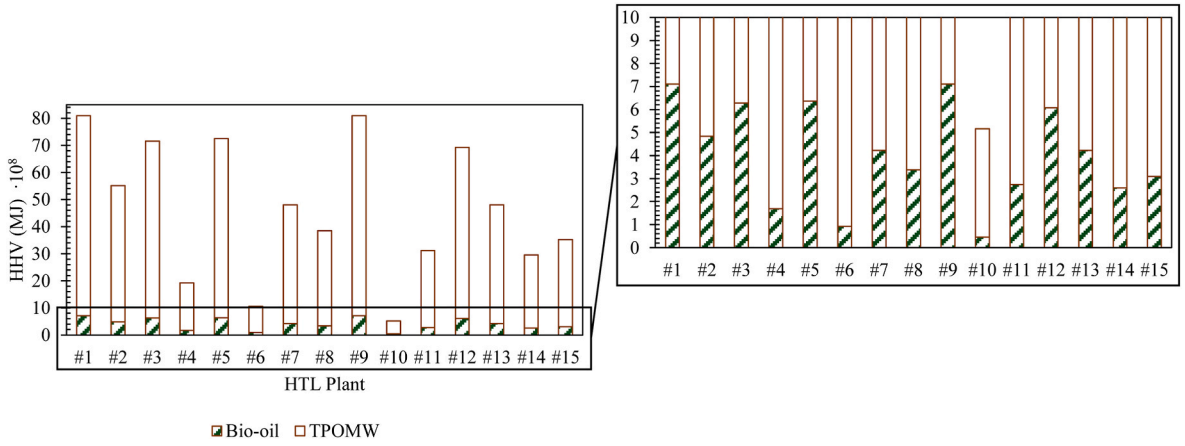



**Fig. 12.** Bio-oil energy potential relative to TPOMW processing capacity per year across the proposed HTL network.

analysis, Data curation. **G. Blázquez:** Validation, Methodology, Investigation, Data curation. **C. Moliner:** Visualization, Supervision, Funding acquisition, Formal analysis, Conceptualization. **E. Aratto:** Visualization, Supervision, Project administration, Funding acquisition. **P. Haro:** Visualization, Supervision, Project administration, Funding acquisition, Conceptualization.

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## Data availability

Data will be made available on request.

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